

Assam Flood Intelligence Platform

Problem Statement & Solution

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Assam's floods cause devastating losses not from lack of response, but because response is **reactive, fragile, and exclusionary**.

- **No early warning of *where* flooding will strike.** River levels, rainfall data, and elevation maps exist in disconnected systems, never combined to predict which villages will flood next — losing the critical 24–48 hour window to act.
- **Alert systems fail exactly when needed most.** Towers go down or congest during peak floods, silencing app-based warnings. Relief camps are sometimes placed on land that later floods too, due to outdated elevation data.
- **Two vulnerable groups are left out entirely.** Farmers get no time-sensitive advisory to harvest or evacuate livestock before water hits. Non-smartphone and low-literacy residents are excluded from app-based alerts altogether.

Solution

A proactive flood-intelligence platform that predicts flooding 48 hours ahead, delivers warnings through any signal condition, and protects both lives and livelihoods.

1. 48-Hour Predictive Engine

Combines river levels, rainfall forecasts, and elevation data (ISRO Bhuvan / NASA SRTM) to predict which villages will flood before it happens.

2. Dynamic Safe-Zone Placement

Continuously recalculates the safest high-ground sites for camps and stockpiles as conditions change.

3. Survival Mode

Auto-downgrades to plain-text alerts on weak 2G signal, and queues messages to fire the instant any connection returns.

4. Crop Protection & Recovery Layer

Sends time-sensitive agricultural advisories before the flood, and allows farmers to upload photos of their fields post-flood for instant AI-driven damage assessment and recovery advice.

5. Universal Two-Way Accessibility

SMS and Assamese IVR voice alerts ensure non-smartphone and low-literacy residents aren't left out, allowing them to reply via basic SMS to ping their SOS location directly to the government dashboard.

6. AI Query Interface

Allows government officials to ask the dashboard questions in natural language (e.g., "Which villages are in the 48-hour flood zone?") for instant, conversational intelligence.

Why it wins: Instead of solving one piece (a chatbot, a map, an alert app), this closes all three major gaps — prediction, delivery, and inclusion — in one connected system.